RISHI WALOKAR

Computer Science & Engineering | AI/ML | Computer Vision

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EDUCATION

St. Vincent Pallotti College of Engineering & Technology, Nagpur | B.Tech. in Computer Science & Engi
CGPA: 8.0/10

ACHIEVEMENTS

- 1st Place - Serve Smart Hackathon, IIT BHU - Developed Trinetra, a YOLO-based military object detection system.
- 1st Place - S.H.E.L.L. Capture The Flag (CTF), VNIT Nagpur - Team SHER.
- Finalist - BYPASS CTF, AIT Pune.
- Smart India Hackathon (SIH) - College-Level Selection, 2x.

SELECTED TECHNICAL PROJECTS

Semantic Retrieval & Multi-Temporal Change Analysis of Satellite Imagery - SIH 26227 | RemoteCLIP - Sentinel-2 - Pytho
- Developed a semantic retrieval pipeline for satellite imagery using RemoteCLIP ViT-B/32.
- Built a local semantic index containing 867 satellite-image patches for natural-language image retrieval.
- Implemented multi-temporal change analysis and integrated backend APIs into a web-based prototype.

Trinetra - Military Object Detection System | YOLO - Computer Vision - Python
- Developed a 12-class object detection system for military and civilian objects using YOLO.
- Worked on dataset preparation, model training, augmentation and evaluation using Precision, Recall, IoU and mAP.
- Developed for the Serve Smart Hackathon at IIT BHU, securing 1st Place.

Offline Tiger Detection System | YOLO - Computer Vision - Python
- Developed an offline wildlife detection system for identifying tigers from images and video without continuous internet connectivity.
- Applied object-detection techniques toward practical wildlife monitoring.

Jeopardy-Style CTF Platform | Web Development - Cybersecurity - JavaScript
- Built a web-based Jeopardy-style CTF platform with challenge categories, scoring and progression mechanisms.
- Designed an interactive competition workflow combining technical challenges with gamification.

Web Scraper | Python - Web Scraping
- Developed a Python-based scraper for automated extraction and processing of web data.

TECHNICAL SKILLS

Programming: Python, C/C++, JavaScript
AI/ML: Machine Learning, Deep Learning, Computer Vision, CNNs, Random Forest, Object Detection, Transfer Learning
Computer Vision: YOLO, Image Classification, IoU, mAP, Data Augmentation
Modern AI: Semantic Retrieval, Embeddings, RemoteCLIP, RAG, LLM Fundamentals
Development: React, Next.js, Flask, Streamlit, REST APIs, Supabase
Cybersecurity & Tools: CTFs, Web Security Fundamentals, Git, GitHub, VS Code, Ultralytics

LEADERSHIP & ACTIVITIES

- Event Lead - Infinity, SVPCET: Led planning and execution of a department-level technical event and coordinated team members and event activities.
- Member - Technex, SVPCET: Contributed to technical activities and student-led initiatives.
- Member - E-Cell, SVPCET: Participated in entrepreneurship-oriented activities and student initiatives.

AREAS OF INTEREST

Artificial Intelligence & Machine Learning | Computer Vision | Remote Sensing | Generative AI | Cybersecurity | Information Retrieval